

Evaluating Recovery of a Dale-Constrained Connectivity Matrix from Synthetic Spike Data: Study Design and Comparison Metrics

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Abstract

We define a protocol for quantifying how accurately the shared connectivity matrix A_{true} of a Dale-law-constrained recurrent Poisson GLM is recovered by PRISM-EM from non-stationary spike trains, as a function of recording duration. Ground truth is known by construction. The evaluation is split into three independent concerns—*support* (which edges exist), *sign* (excitatory vs. inhibitory), and *magnitude* (edge weight)—of which this document treats support and magnitude in detail. We fix the pruning threshold minW and the regularization strength for the present study, apply minW to the raw fitted weights *before* any spectral rescaling, and compare magnitudes only after normalizing both matrices to unit spectral radius. We motivate the expected $\propto 1/\sqrt{n_{\text{eff}}}$ scaling of weight estimation error with effective sample size and use it as the quantitative null hypothesis for the duration sweep.

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1 Background: Data and Fit in Brief

1.1 Generator

A network of N neurons (N_E excitatory, $N_I = N - N_E$ inhibitory) is simulated as a lag-1 Poisson vector autoregression (VAR(1)). A single shared connectivity matrix $A_{\text{true}} \in \mathbb{R}^{N \times N}$ governs all dynamics; M latent states differ only in a per-state bias (log-baseline) vector B_m . At each time bin t of width Δt ,

$$\eta_t = A_{\text{true}} Y_{t-1} + B_{\text{eff},t}, \quad \lambda_t = \exp(\min(\eta_t, \eta_{\text{clip}})) \Delta t, \quad Y_t \sim \text{Poisson}(\lambda_t), \quad (1)$$

where $B_{\text{eff},t} = \sum_m c_{t,m} B_m$ mixes the state biases along a smooth simplex trajectory c_t . Column convention: $(A_{\text{true}})_{ij}$ is the weight *from* presynaptic neuron j *to* postsynaptic neuron i . Dale’s law is enforced at the source: every outgoing weight of an excitatory neuron is positive, every outgoing weight of an inhibitory neuron is negative, and self-loops are negative. The binary support $E_{\text{true}} \in \{0, 1\}^{N \times N}$ is generated by per-neuron i.i.d. Bernoulli connectivity, and the spectral radius is fixed at $\rho(A_{\text{true}}) = R < 1$. Ground-truth arrays $(A_{\text{true}}, \{B_m\}, E_{\text{true}})$ and the (x, y) coordinates of all neurons are saved.

1.2 Fitter (PRISM-EM)

PRISM-EM recovers $\hat{A}$ and $\{\hat{B}_m\}$ by block coordinate descent: an E-step infers the simplex coefficients $c_{1:T}$ and Viterbi-decodes them to one-hot state assignments; an M-step updates $\hat{A}, \hat{B}$ by Adam on the Poisson negative log-likelihood with off-diagonal ℓ_1 shrinkage (strength λ_3) and a spectral-radius projection to ρ_{max} . The fitter does *not* impose Dale’s law, so recovered rows may contain mixed signs; this is exploited later only as a sign-consistency diagnostic and is out of scope here. Because A_{true} is shared across all states, every time bin contributes to estimating $\hat{A}$, and the arbitrary permutation of recovered state labels does *not* affect $\hat{A}$; the connectivity comparison therefore requires no label alignment.

2 Part I: Study Design and Conventions

2.1 Operating Regime

We target $N \in [50, 500]$ neurons with a median single-neuron firing rate of 2–3 Hz (range $[1, 5]$ Hz), spikes binned at $\Delta t = 10$ ms, and recording durations from 10 to 60 minutes. The latent dynamics use $M \in \{2, 3\}$ states with mean dwell time ~ 1 s. The central scientific question is: *how much recovery quality is lost by using a 10-minute record instead of a 60-minute record?*

2.2 Master-Record Subsampling

To remove generator variability from the duration comparison, a single *master* record of 120 minutes is generated per configuration (N, seed) . Shorter records are obtained as contiguous temporal prefixes,

$$Y^{(D)} = Y_{1:T_D}, \quad T_D = \lfloor D/\Delta t \rfloor, \quad (2)$$

for durations $D \in \{10, 15, 20, 30, 45, 60\}$ minutes, with the full 120-minute record retained as an asymptotic ceiling. Contiguous prefixes (rather than random bin subsets) preserve the temporal lag structure and the state-dwell statistics that the fitter depends on. Each metric is reported relative to its 120-minute value, so results read as “fraction of asymptotic recovery retained.” At least 5 independent seeds are run per (N, D) pair; sampling variability in this sparse, low-rate regime is large and single-seed results are not interpretable.

2.3 Parameters Held Constant (Present Study)

For this phase the following are fixed across all durations and seeds:

- Pruning threshold $\text{minW} = 0.02$ (applied to raw $\hat{A}$; see below).
- ℓ_1 strength λ_3 and all other fit hyperparameters (learning rates, EM/M-step iteration counts, $\rho_{\max}$, delay schedules).
- Network structure $(N, N_E, R, E_{\text{true}}, A_{\text{true}}, \{B_m\})$ within a seed.

Holding λ_3 fixed means we measure *fixed-regularizer* recovery. Because the optimal ℓ_1 penalty scales with the noise level ($\propto 1/\sqrt{n_{\text{eff}}}$, Section 2.5), a fixed λ_3 over-shrinks short records; the measured duration effect therefore includes a regularization component that we acknowledge explicitly and revisit in later work.

2.4 Threshold Then Rescale: Order of Operations

Support and magnitude use different normalizations of $\hat{A}$, and the order matters.

1. **Support** is evaluated on the *raw* fitted matrix. An edge (i, j) is declared present iff $|\hat{A}_{ij}| > \text{minW}$, $i \neq j$. The threshold is absolute and is applied at the fitter’s native scale ($\rho \approx \rho_{\max}$).
2. **Magnitude** is evaluated after rescaling both matrices to unit spectral radius,

$$\hat{A}^* = \hat{A}/\rho(\hat{A}), \quad A_{\text{true}}^* = A_{\text{true}}/\rho(A_{\text{true}}), \quad (3)$$

which removes the global scale mismatch between the fitter’s $\rho_{\max}$ constraint and the generator’s R . This rescaling is a single scalar per matrix; it preserves all signs, the zero pattern, and all relative orderings.

The two normalizations are kept in separate pipelines. Applying the absolute minW after a ρ -rescaling would make its effective cut size depend on N (since unit- ρ entries scale roughly as the inverse square root of the number of active edges), introducing an N -dependent artifact into the support results; thresholding the raw weights avoids this. All metrics, both support and magnitude, exclude the diagonal.

2.5 Effective Sample Size and the $1/\sqrt{n_{\text{eff}}}$ Law

The recoverability of an edge $j \rightarrow i$ is governed not by raw duration but by how often the presynaptic neuron j is observed firing. The effective sample size for that edge is approximately the number of bins in which j is active,

$$n_{\text{eff}}(j) \approx f_j T_{\text{rec}}, \quad (4)$$

with f_j the firing rate of neuron j and T_{rec} the record duration. At the median rate $f_j \approx 2.5$ Hz this gives $n_{\text{eff}} \approx 1,500$ presynaptic spikes at 10 minutes and $\approx 9,000$ at 60 minutes; for the slowest neurons (1 Hz) the 10-minute count drops to ≈ 600 . The slowest presynaptic neurons therefore bound the worst-case edge recovery.

For maximum-likelihood estimation of GLM coefficients, the standard error of each weight scales as

$$\text{SE}(\hat{A}_{ij}) \propto \frac{1}{\sqrt{n_{\text{eff}}(j)}}. \quad (5)$$

Reducing the record from 60 to 10 minutes is a $6\times$ reduction in data, predicting

$$\frac{\text{SE}_{10}}{\text{SE}_{60}} \approx \sqrt{6} \approx 2.45. \quad (6)$$

This is the quantitative *null hypothesis* for the duration sweep: magnitude-estimation error should grow by $\approx 2.45\times$ from 60 to 10 minutes if sampling noise alone is responsible. Larger growth indicates

an additional failure mode (e.g. EM converging to a worse optimum, or support errors corrupting the surviving magnitudes); smaller growth indicates a regularization floor where λ_3 dominates sampling noise and duration matters little. **Important caveat:** support recovery does *not* follow this smooth law. Weak edges sit near a detection threshold; as n_{eff} falls, edges near that threshold drop out abruptly, so recall (especially of weak edges) degrades in a step-like fashion while precision can collapse from proliferating noise-floor false positives.

3 Part II: Comparing Support

The predicted support is the off-diagonal binary mask $\hat{E}_{ij} = \mathbf{1}\{|\hat{A}_{ij}| > \text{minW}\}$; ground truth is E_{true} . On the off-diagonal entries define the confusion counts

$$\text{TP, FP, FN, TN} \quad (7)$$

(edge present in both / only in $\hat{E}$ / only in E_{true} / absent in both). Because the network is sparse ($\sim 5\text{--}20\%$ of entries are edges), the overwhelming majority of entries are TN, and any metric that rewards TN will read optimistically. We therefore favor edge-focused metrics and always report the raw counts (TP, FP, FN) so that a change in any summary can be attributed to missed edges (FN) versus hallucinated edges (FP), which are distinct failure modes.

3.1 Candidate Scalar Metrics

All are evaluated at the fixed $\text{minW} = 0.02$.

Jaccard index (primary).

$$J = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}. \quad (8)$$

Ignores TN entirely; answers “of all edges either matrix claims, what fraction do both agree on.” Most conservative and most edge-focused of the candidates.

F_1 score.

$$F_1 = \frac{2 \text{TP}}{2 \text{TP} + \text{FP} + \text{FN}}. \quad (9)$$

Also ignores TN. F_1 and J are monotone transforms of one another, $J = F_1/(2 - F_1)$, so they rank fits identically; J is simply harsher on the same scale. Retained for comparison only.

Precision and Recall.

$$\text{Prec} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \quad \text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \quad (10)$$

Reported as the decomposition underlying J and F_1 : recall tracks loss of real edges as n_{eff} falls, precision tracks the proliferation of noise-floor false positives.

Matthews correlation coefficient (secondary).

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}}. \quad (11)$$

Uses all four cells, including TN, so it reads higher than J in the sparse regime; retained for comparability with the broader literature. Reports the overall correlation between the two binary matrices.

We will run all of the above initially and discard the least discriminating once empirical sensitivity is known.

3.2 Block-Resolved Support (E/I structure)

The neuron ordering induces four meaningful blocks, indexed by presynaptic type (*column* block, by the from- j convention) and postsynaptic type (*row* block):

$$E \rightarrow E, \quad E \rightarrow I, \quad I \rightarrow E, \quad I \rightarrow I. \quad (12)$$

Each scalar metric of Section 3 is recomputed within each block. Because inhibitory neurons are a minority ($\sim 20\%$) and typically carry different effective drive, we expect $I \rightarrow \cdot$ edges to recover more poorly; the block table makes this circuit-meaningful rather than arbitrary.

3.3 Spatially Coarse-Grained Support

Using the known (x, y) coordinates, overlay a coarse $G \times G$ grid on the neuron positions. Let $g(i) \in \{1, \dots, G^2\}$ be the grid cell of neuron i , and partition neurons additionally by type $\tau(i) \in \{E, I\}$. Define a *lumped* (super-node) connectivity between ordered cell-type pairs by counting only edges that leave one cell and enter another. For source cell a of type σ and target cell b of type (any, or fixed), the lumped true and recovered edge counts are

$$\mathcal{N}_{(a,\sigma) \rightarrow b}^{\text{true}} = \sum_{\substack{j: g(j)=a, \tau(j)=\sigma \\ i: g(i)=b, i \neq j}} (E_{\text{true}})_{ij}, \quad (13)$$

$$\mathcal{N}_{(a,\sigma) \rightarrow b}^{\text{hat}} = \sum_{\substack{j: g(j)=a, \tau(j)=\sigma \\ i: g(i)=b, i \neq j}} \mathbf{1}\{|\hat{A}_{ij}| > \text{minW}\}. \quad (14)$$

Restricting the source sum to a single type σ keeps excitatory and inhibitory super-edges separate, as intended. Comparison at the super-node level can use either a count-agreement measure,

$$\text{err}_{(a,\sigma) \rightarrow b} = \frac{|\mathcal{N}_{(a,\sigma) \rightarrow b}^{\text{hat}} - \mathcal{N}_{(a,\sigma) \rightarrow b}^{\text{true}}|}{\mathcal{N}_{(a,\sigma) \rightarrow b}^{\text{true}} + 1}, \quad (15)$$

or a fine-grained Jaccard restricted to the edge set between the two cells. Rendering $\text{err}_{(a,\sigma) \rightarrow b}$ as a spatial heatmap reveals whether recovery quality varies across the tissue. **Interpretive caution:** in the current generator the binary support is spatially i.i.d., so E_{true} has no intrinsic spatial structure. Any spatial pattern in recovery is therefore an *inference artifact*, and is expected to be explained almost entirely by the spatial distribution of presynaptic firing rates rather than by position per se (Section 3.4). The grid view is thus a descriptive lens; the mechanistic axis is firing rate.

3.4 Stratified Recovery

The mechanistic drivers of edge recoverability are presynaptic firing rate and true edge magnitude. We therefore stratify recall:

- by presynaptic rate f_j (binned),
- by true magnitude $|A_{\text{true}}_{ij}|$ (binned),
- jointly, as a two-dimensional recall heatmap over $(f_j, |A_{\text{true}}_{ij}|)$.

The joint heatmap is the single most explanatory support diagnostic: it shows exactly which edges become unrecoverable as duration shrinks, and it should account for any apparent spatial or block-wise pattern.

4 Part III: Comparing Magnitude

Magnitude comparison is restricted to the true support ($E_{true_{ij}} = 1, i \neq j$) and uses the unit- ρ matrices $Ahat^*, Atrue^*$ of Section 2.4. False-positive edges are the concern of the support metrics and are excluded here, except where noted (the global-vs-support Frobenius gap below). Because the true edges span both signs (positive E, negative I), a metric computed across all signs is partly rewarded for merely separating E from I—the easy part. To isolate magnitude fidelity from sign, the primary forms of each metric are computed *within* the same-sign edge sets (within-E and within-I), with the global (both-sign) form reported as a secondary, sign-inflated reference.

4.1 Rank Fidelity: Spearman Correlation

Over the chosen edge set, the Spearman correlation ρ_S between $\{Atrue^*_{ij}\}$ and $\{Ahat^*_{ij}\}$ measures preservation of the *ordering* of edge strengths:

$$\rho_S = \text{corr}(\text{rank}(Atrue^*), \text{rank}(Ahat^*)). \quad (16)$$

Computed within-E, within-I, and globally. Spearman is invariant to any monotone rescaling, so it is unaffected by the unit- ρ normalization—a useful internal consistency check (a rescaling bug moves Pearson but leaves Spearman fixed). It weights all ranks equally and is therefore sensitive to the fidelity of *weak* edges.

4.2 Magnitude Error: Relative Frobenius Norm

Two complementary versions are reported.

$$\text{RFE}_{\text{supp}} = \frac{\|(Ahat^* - Atrue^*) \odot E_{\text{true}}\|_F}{\|Atrue^* \odot E_{\text{true}}\|_F}, \quad (17)$$

$$\text{RFE}_{\text{glob}} = \frac{\|Ahat^* - Atrue^*\|_F}{\|Atrue^*\|_F}, \quad (18)$$

both over off-diagonal entries. RFE_{supp} measures magnitude error on real edges; RFE_{glob} additionally includes the mass placed on false-positive entries. The gap $\text{RFE}_{\text{glob}} - \text{RFE}_{\text{supp}}$ is a single-number summary of spurious recovered mass and is expected to grow as duration shrinks. The Frobenius norm is dominated by the largest-magnitude edges and is thus sensitive to the *strong* edges—complementary to Spearman.

4.3 Systematic Shrinkage: Regression Slope

Fit, over the chosen edge set, the linear relation

$$Ahat^*_{ij} = \beta Atrue^*_{ij} + \gamma, \quad (19)$$

reporting the slope β and intercept γ within-E and within-I, plus a zero-intercept variant ($\gamma \equiv 0$). The slope β is the shrinkage diagnostic. After unit- ρ rescaling the spectral-constraint mismatch is removed, so a residual $\beta \neq 1$ is attributable to ℓ_1 shrinkage and finite-sample estimation bias; ℓ_1 is expected to push $\beta < 1$, more so for the noisier inhibitory edges. A nonzero γ on the support-restricted fit flags a magnitude offset, which is physically unexpected (no edge implies no systematic shift).

4.4 The Underlying Scatter

The three scalars above are summaries of one object: the scatter of $Ahat_{ij}^*$ against $Atrue_{ij}^*$ over the true support, colored by presynaptic rate f_j . This plot reveals shrinkage (slope), heteroscedasticity (weak/slow edges noisier), and outliers (slow-presynaptic edges) at a glance, and should be inspected before interpreting the scalars.

4.5 Sensitivity Complementarity (Summary)

The three magnitude metrics are deliberately complementary in what they emphasize:

Metric	Emphasizes	Scale dependence
Spearman ρ_S	ordering of all edges (incl. weak)	invariant
Relative Frobenius	large-magnitude (strong) edges	requires unit- ρ
Regression slope β	systematic shrinkage bias	requires unit- ρ

5 Headline Deliverable

The duration study is summarized by curves of the retained primary metrics

$$\{J, \rho_S^E, \rho_S^I, \text{RFE}_{\text{supp}}, \beta^E, \beta^I\} \quad \text{vs.} \quad D \in \{10, \dots, 60, 120\} \text{ min}, \quad (20)$$

each normalized to its 120-minute value and averaged over seeds with error bars. The magnitude-error curve is overlaid with the $\sqrt{6} \approx 2.45$ sampling-noise reference (Section 2.5); deviations from this reference, together with the step-like behavior of recall in the stratified heatmap (Section 3.4), constitute the principal findings on information loss between the 10- and 60-minute records.